

SOP 102: Medical Emergencies

1 Summary

This document establishes procedures for medical emergencies at the Notre Dame Human Neuroimaging Center (ND-HNC) MRI scanner. They are intended to coordinate a planned response of:

1. Moving to safety,
2. Initiating emergency response,
3. Providing First Aid,
4. Coordinating with others in the building,
5. Protecting responders from the magnetic field.

For dual responders, the Lead protects the participant and then equipment while the Helper initiates the emergency response.

2 Scope

These policies apply to all investigators, research assistants, and ND-HNC staff who enter Zones 3 & 4 (B15, B17, B17a; Form RO03).

3 Definitions

- **Control Room:** B17; The main room of the MRI suite that contains the computer displays and keyboards that control the MRI scanner. The Control Room is accessed from B15 that is controlled by an Electronic Lock.
- **Electronic Lock:** All doors at the ND-HNC are controlled by electronic locks. The electronic lock system recognizes the owner's Irish1Card, logs the time of the attempted entry, and unlocks the door if the owner has been granted access.
- **Helper:** Second person in Zones 3 and 4 who assists in emergency responses, typically Level 2-3.
- **Lead:** Primary person in Zones 3 and 4 responsible for the Visitor's safety, in charge during emergency responses. Level 4 operator or MRI Technologist.

- **Magnet Room:** B17A; The room within the MRI suite that contains the MRI scanner. It is accessed from within the MRI Control room (B17) through a doorway that is controlled by an Electronic Lock.
- **Visitor:** An individual who has not taken the ND-HNC MRI Safety Course but has a legitimate reason for entering the MRI suites. Research participants and their legal guardians are Visitors. Other Visitors may include: individuals providing/receiving a tour of the VFPC and ND-HNC, faculty and staff who have not received MRI Safety Course, and research assistants.

4 Notes

1. Emergency response and maintenance personnel **CANNOT** enter the Magnet Room unless they have been screened and all metal objects have been removed.
2. Emergency response personnel can **ONLY** enter Zone 3 without completing an MRI Safety screener if the door to the Magnet Room has been locked by key.
3. The First-Aid kit and Automated External Defibrillator (AED) are **NOT** MRI compatible.

5 Policies & Procedures

In the sections below, several scenarios are depicted regarding medical emergencies.

5.1 Injury: a CONSCIOUS individual sustains an injury requiring medical attention.

Given the volume of staff, researchers, and participants at the ND-HNC, there is a realistic possibility that an accident or medical emergency will occur. Accordingly, the center must maintain a constant state of readiness to respond. Ensuring safety means having clear medical protocols in place so that any incident is handled swiftly and effectively.

Single Responder

1. Escort them to B15 (Form RO03).
2. Call 911 and say: “We have a person that needs medical attention in the MRI Center at Veldman Family Psychology Clinic, 501 North Hill Street”.
3. Call Veldman Reception and say “We have a medical emergency and have called 911.”
4. Retrieve first-aid kit, administer first aid [TODO PIC].

5. Close and lock Magnet Room door.
6. Prepare to receive and escort emergency medical services.
7. Call any MRI staff.
8. Record Adverse Event in ND-HNC Form LG03: Incidents.

Dual Responders

1. Lead
 - 1.1 Escort them to B15 (Form RO03).
 - 1.2 Retrieve first-aid kit, administer first aid [TODO PIC].
 - 1.3 Close and lock Magnet Room door.
 - 1.4 Prepare to receive and escort emergency medical services.
 - 1.5 Call any MRI staff.
 - 1.6 Record Adverse Event in ND-HNC Form LG03: Incidents.
2. Helper
 - 2.1 Call 911 and say: “We have a person that needs medical attention in the MRI Center at Veldman Family Psychology Clinic, 501 North Hill Street”.
 - 2.2 Call Veldman Reception and say “We have a medical emergency and have called 911.”
 - 2.3 Assist in first aid.
 - 2.4 Assist receiving and escorting emergency medical services.

5.2 Injury: an UNCONSCIOUS individual sustains an injury requiring medical attention.

As above, an increased number of individuals in the MRI suite corresponds with an increased risk of an injury. Further, significant injury and/or hardware malfunction (e.g. quench) may render an individual unconscious.

Single Responder

1. **Unconscious individual in Control Room:** move them into the recovery position.
2. **Unconscious individual in Magnet Room:** use MR compatible gurney to move them to B15 [TODO PIC].

3. Call 911 and say: “We have a person that needs medical attention in the MRI Center at Veldman Family Psychology Clinic, 501 North Hill Street”.
4. Call Veldman Reception and say “We have a medical emergency and have called 911.”
5. Retrieve first-aid kit, administer first aid [TODO PIC].
6. Lock Magnet Room door.
7. Prepare to receive and escort emergency medical services.
8. Call any MRI staff.
9. Record Adverse Event in ND-HNC Form LG03: Incidents.

Dual Responders

1. Lead
 - 1.1 **Unconscious individual in Control Room:** move them into the recovery position.
 - 1.2 **Unconscious individual in Magnet Room:** use MR compatible gurney to move them to B15 [TODO PIC].
 - 1.3 Retrieve first-aid kit, administer first aid [TODO PIC].
 - 1.4 Close and lock Magnet Room door.
 - 1.5 Stay with individual, continue to administer aid.
 - 1.6 Call any MRI staff.
 - 1.7 Record Adverse Event in ND-HNC Form LG03: Incidents.
2. Helper
 - 2.1 Call 911 and say: “We have a person that needs medical attention in the MRI Center at Veldman Family Psychology Clinic, 501 North Hill Street”.
 - 2.2 Call Veldman Reception and say “We have a medical emergency and have called 911.”
 - 2.3 Assist in first aid.
 - 2.4 Prepare to assist and escort emergency medical services.

5.3 Chest and/or arm pain, respiration difficulty, loss of consciousness

When conducting research with adults and older adults, certain procedures or stressors may inadvertently trigger symptoms of a heart attack. It is absolutely critical that we take any symptoms seriously and act immediately. Prioritizing participant safety means being vigilant, recognizing the warning signs early, and avoiding hesitation when a situation demands urgent medical intervention.

Single Responder

1. **Conscious individual:** escort them to B15 (Form RO03).
2. **Unconscious individual in Control Room:** move them into the recovery position.
3. **Unconscious individual in Magnet Room:** use MR compatible gurney to move them to B15 [TODO PIC].
4. Call 911 and say: “We have a person with [chest pain, respiratory problems, unresponsiveness] that needs medical attention in the MRI Center at Veldman Family Psychology Clinic, 501 North Hill Street”.
5. Call Veldman Reception and say “We have a medical emergency and have called 911.”
6. Retrieve Automatic External Defibrillator (AED), prepare to administer aid [TODO PIC].
7. Close and lock Magnet Room door.
8. Prepare to receive and escort emergency medical services.
9. Call any MRI staff.
10. Record Adverse Event in ND-HNC Form LG03: Incidents.

Dual Responders

1. Lead
 - 1.1 **Conscious individual:** escort them to B15 (Form RO03).
 - 1.2 **Unconscious individual in Control Room:** move them into the recovery position.
 - 1.3 **Unconscious individual in Magnet Room:** use MR compatible gurney to move them to B15 [TODO PIC].
 - 1.4 Retrieve Automatic External Defibrillator (AED), prepare to administer aid [TODO PIC].

- 1.5 Lock Magnet Room door.
- 1.6 Stay with individual, continue to administer aid.
- 1.7 Call any MRI staff.
- 1.8 Record Adverse Event in ND-HNC Form LG03: Incidents.

2. Helper

- 2.1 Call 911 and say: “We have a person with [chest pain, respiratory problems, unresponsiveness] that needs medical attention in the MRI Center at Veldman Family Psychology Clinic, 501 North Hill Street”.
- 2.2 Call Veldman Reception and say “We have a medical emergency and have called 911.”
- 2.3 Assist in first aid.
- 2.4 Prepare to assist and escort emergency medical services.

5.4 Metal: pinned Visitor WITHOUT risk to life or limb

Inexperience, complacency, and/or over-confidence when entering the Magnet Room can inadvertently introduce ferromagnetic materials into the magnetic field. When a Visitor is present, they may be pinned to the bore by the materials. This SOP details what to do if there is not any risk to the Visitor’s life or limb i.e. an injury has not occurred or is very minor.

Single Responder

1. Do not attempt to remove the item.
2. Visitor is injured: provide first aid.
3. Call any MRI staff; MRI staff call Richard Vega (Siemens Engineer) to determine if the magnet must be quenched.
4. Record Adverse Event in ND-HNC Form LG03: Incidents.
5. Record compliance issue in ND-HNC Form LG04: Compliance.

Dual Responders

1. Lead
 - 1.1 Do not attempt to remove the item.
 - 1.2 Call any MRI staff; MRI staff call Richard Vega (Siemens Engineer) to determine if the magnet must be quenched.

1.3 Record Adverse Event in ND-HNC Form LG03: Incidents.

1.4 Record compliance issue in ND-HNC Form LG04: Compliance.

2. Helper

2.1 Visitor is injured: provide first aid.

5.5 Metal: pinned Visitor WITH risk to life or limb

Inexperience, complacency, and/or over-confidence when entering the Magnet Room can inadvertently introduce ferromagnetic materials into the magnetic field. When a Visitor is present, they may be pinned to the bore by the materials. This SOP details what to do when risks to the Visitor's life or limb exists i.e. injury has occurred, or is likely to occur.

Single Responder

1. Quench the magnet (Fig. 1; Fig. 2, A).

1.1 A quench cannot be stopped once it is initiated.

1.2 As the helium boils off, the magnet will decrease in strength.

1.3 It will take approximately 3 minutes for the magnet to deactivate.

2. Call 911 and say: "We have a person that needs medical attention in the MRI Center at Veldman Family Psychology Clinic, 501 North Hill Street".

3. Call Veldman Reception and say "We have a medical emergency and have called 911."

4. Retrieve First-Aid kit.

5. If gas fails to vent, once magnet is ramped down use MRI gurney and follow SOP 101.5.1.

6. Administer first aid.

7. Prepare to receive and escort emergency medical services.

8. Call any MRI staff.

9. Record Adverse Event in ND-HNC Form LG03: Incidents.

10. Record compliance issue in ND-HNC Form LG04: Compliance.

Dual Responders

1. Lead
 - 1.1 Quench the magnet (Fig. 1; Fig. 2, A).
 - 1.1.1 A quench cannot be stopped once it is initiated.
 - 1.1.2 As the helium boils off, the magnet will decrease in strength.
 - 1.1.3 It will take approximately 3 minutes for the magnet to deactivate.
 - 1.2 If gas fails to vent, once magnet is ramped down use MRI gurney and follow SOP 101.5.1.
 - 1.3 Administer first aid.
 - 1.4 Call any MRI staff.
 - 1.5 Record Adverse Event in ND-HNC Form LG03: Incidents.
 - 1.6 Record compliance issue in ND-HNC Form LG04: Compliance.
2. Helper
 - 2.1 Call 911 and say: “We have a person that needs medical attention in the MRI Center at Veldman Family Psychology Clinic, 501 North Hill Street”.
 - 2.2 Call Veldman Reception and say “We have a medical emergency and have called 911.”
 - 2.3 Retrieve First-Aid kit.
 - 2.4 Assist in administering first aid.
 - 2.5 Prepare to receive, escort emergency medical services.

5.6 Seizure or stroke

Visitors may have a complex medical history, and we need to be vigilant for, and responsive to, any signs of medical distress.

1. Seizure
 - 1.1 Do not attempt to hold person still.
 - 1.2 Provide cushioning for head.
 - 1.3 Follow relevant injury procedure (SOP 102.5.5.2)
2. Stroke
 - 2.1 Act F.A.S.T.
 - F: Face - signs of drooping

- A: Arms - arm weakness
- S: Speech - difficulty speaking, slurring
- T: Time - initiate 911 immediately

2.2 Follow relevant injury procedure (SOP 102.5.5.2)

6 Figures



Figure 1: Control Room MRI quench button.



Figure 2: Magnet Room button wall. **A** Quench. **B** Oxygen sensor, alarm. **C** Emergency Power Off.